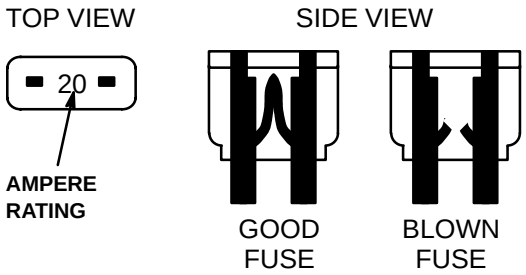
# 11-1 FUSE PANEL/CIRCUIT PROTECTION



## CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fusible links, fuse link cartridges, circuit breakers, or a combination of these devices.

### BLADE TYPE FUSE

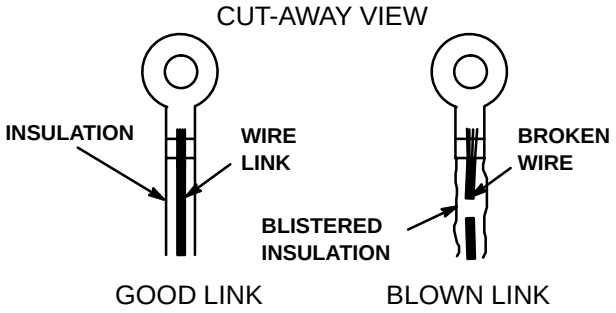



Blade type fuses have a transparent plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code shown here:

| BLADE FUSE COLOR CODING | |
|---|---|
| **AMPERE RATING** | **HOUSING COLOR** |
| 4 | Pink |
| 5 | Tan |
| 10 | Red |
| 15 | Light Blue |
| 20 | Yellow |
| 25 | Natural |
| 30 | Light Green |

### FUSIBLE LINK



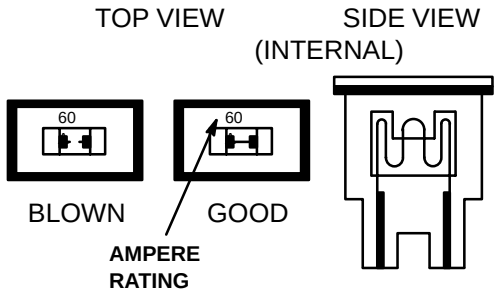

Fusible links are short lengths of wire that are smaller in diameter than the wires they are protecting. Fusible link wire is covered with a special thick, non-flammable insulation. An overload condition causes the insulation to blister. If the overload condition continues, the wire link will melt. To check a fusible link, look for blistered insulation. If the insulation is okay, pull lightly on the wire; If the fusible link stretches, the wire has melted.

When replacing fusible links, first cut the protected wire where it is connected to the fusible link. Then, tightly crimp or solder the new link to the protected wire.

Fusible links are often identified by color coding of the insulation, as shown here:

| FUSIBLE LINK COLOR CODING | |
|---|---|
| **WIRE LINK SIZE** | **INSULATION COLOR** |
| 20 GA | Blue |
| 18 GA | Brown or Red |
| 16 GA | Black or Orange |
| 14 GA | Green |
| 12 GA | Gray |

### FUSE LINK CARTRIDGE




Fuse link cartridges have a colored plastic housing with a clear “window” at the top. To check a fuse link cartridge, look at the fuse element through the clear “window.”

To replace a fuse link cartridge, pull it from the fuse box or panel. High ampere cartridges (80 A or more) are held in place by screws, which must be removed in order to replace the cartridge. Always replace a blown fuse link cartridge with a new one having the same ampere rating.

The ampere rating of a fuse link cartridge can also be determined by following the color code shown here:

| FUSE LINK CARTRIDGE COLOR CODING | |
|---|---|
| **AMPERE RATING** | **HOUSING COLOR** |
| 20 | Light Blue |
| 30 | Pink |
| 40 | Green |
| 50 | Red |
| 60 | Yellow |
| 80 | Black |
| 100 | Dark Blue |

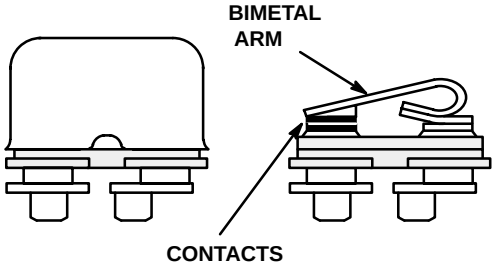
### CIRCUIT BREAKER

Some circuits are protected by circuit breakers (abbreviated “c. b.” in fuse chart).
They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

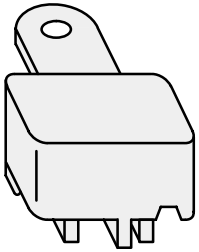
Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other the arm bends, opening the contacts. Current flow is broken. A circuit breaker can be the cycling or non-cycling type.

### FUSE PANEL MOUNTED CYCLING TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)



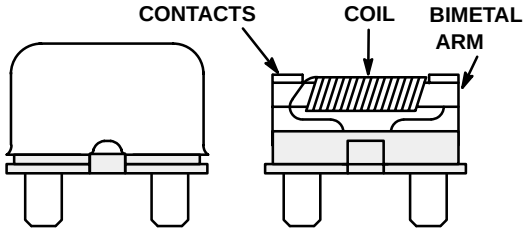

### IN-LINE MOUNTED CYCLING TYPE



In the cycling type, the bimetal arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

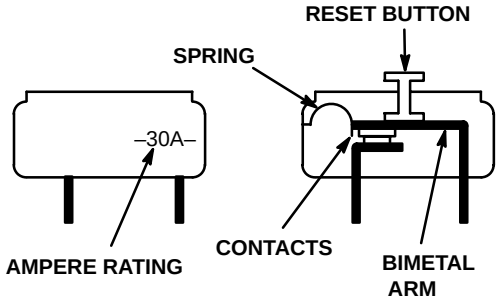
### FUSE PANEL MOUNTED NON-CYCLING TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)




### FUSE PANEL MOUNTED MANUAL RESET TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)

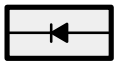



Two types of non-cycling circuit breakers are used; one is reset by removing power from the circuit, and the other is reset by depressing a reset button.

In the first type, there is a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

In the second type, a spring pushes the bimetal arm down and holds the contacts together. When an overcurrent condition exists and the bimetal arm heats up, the bimetal arm bends enough to overcome the spring and the contacts snap open. The contacts stay open until the reset button is pushed and the contacts snap together again.

### DIODE




Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

# 11-3 FUSE PANEL/CIRCUIT PROTECTION

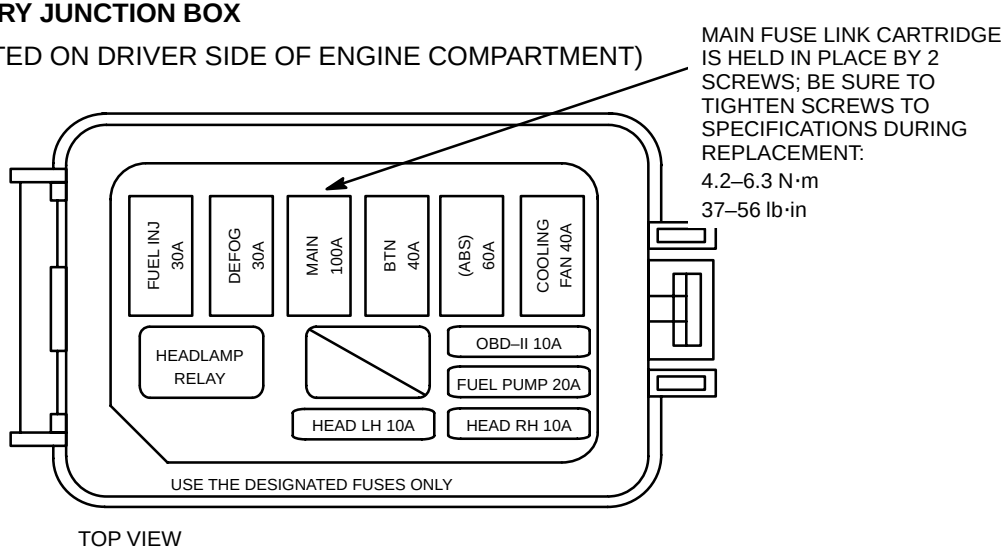


| FUSE LINK CARTRIDGE | AMPS | CIRCUITS PROTECTED |
|---|---|---|
| FUEL INJ. | 30 | AIR BAGS, ENGINE CONTROLS, GENERATOR |
| DEFOG. | 30 | REAR WINDOW DEFROST |
| MAIN | 100 | OVERALL CIRCUIT PROTECTION |
| BTN | 40 | HAZARD STOP, DOOR LOCK, TAIL, ROOM AND HORN FUSES OF THE CENTRAL JUNCTION BOX |
| ABS | 60 | ABS MAIN RELAY |
| COOLING FAN | 40 | CONSTANT CONTROL RELAY MODULE |
| **FUSE** | **AMPS** | **CIRCUITS PROTECTED** |
| OBD-II | 10 | DATA LINK CONNECTOR (DLC), INSTRUMENT CLUSTER |
| FUEL PUMP | 20 | ENGINE CONTROLS |
| HEAD RH | 10 | HEADLAMPS |
| HEAD LH | 10 | HEADLAMPS |

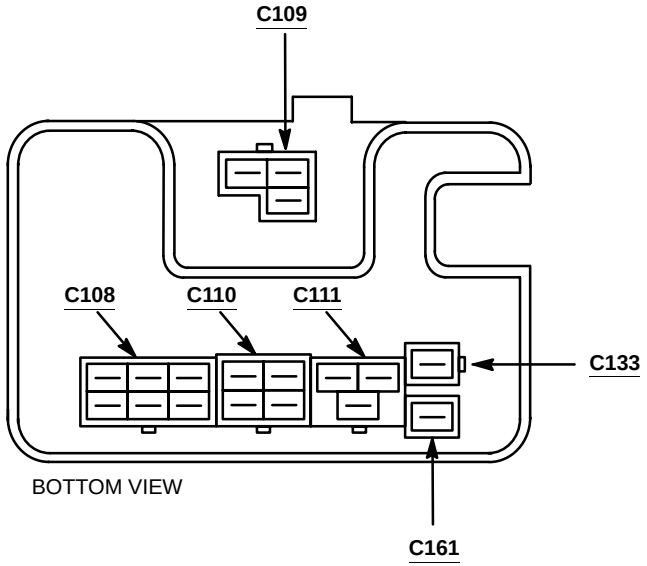
| AMPERE RATING | HOUSING COLOR |
|---|---|
| **BLADE TYPE FUSE** | |
| 10 | Red |
| 15 | Blue |
| 20 | Yellow |
| 30 | Green |
| **FUSE LINK CARTRIDGE** | |
| 20 | Light Blue |
| 30 | Pink |
| 40 | Green |
| 50 | Red |
| 60 | Yellow |
| 80 | Black |
| 100 | Dark Blue |

**BATTERY JUNCTION BOX**

(LOCATED ON DRIVER SIDE OF ENGINE COMPARTMENT)



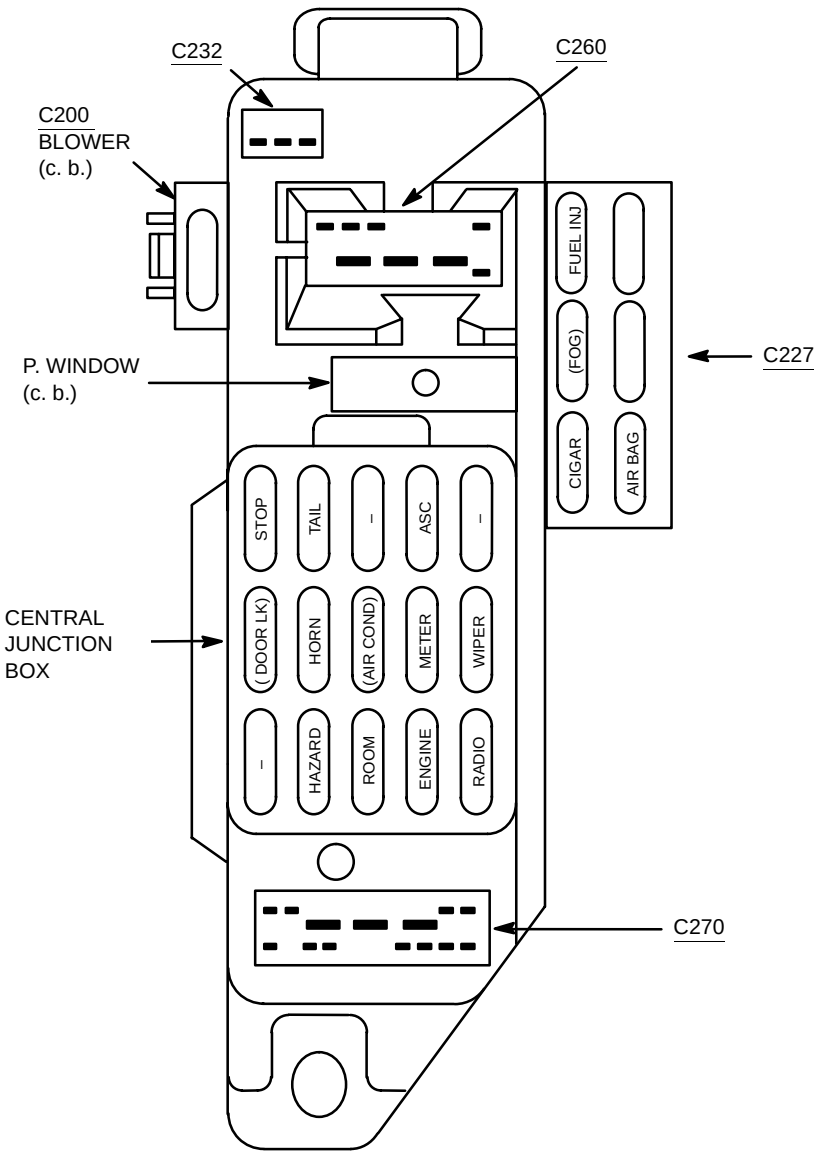

TOP VIEW

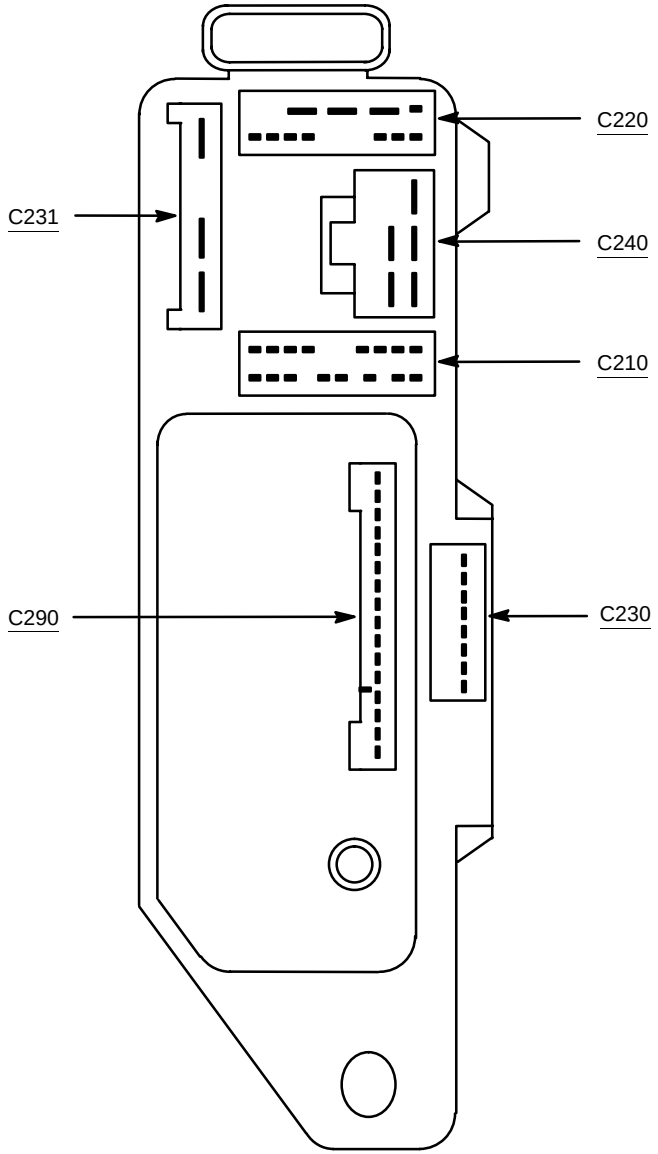



BOTTOM VIEW

**CENTRAL JUNCTION BOX**

(LOCATED ABOVE THE LEFT SIDE KICK PANEL)



# 11-5 FUSE PANEL/CIRCUIT PROTECTION



| FUSE | AMPS | CIRCUITS PROTECTED |
|---|---|---|
| ASC | 10 | SPEED CONTROL |
| BLOWER (C. B.) | 30 | BLOWER MOTOR RELAY |
| CIGAR | 20 | CIGAR LIGHTER |
| ENGINE | 15 | ELECTRONIC AUTOMATIC TRANSAXLE, IGNITION SYSTEM, CONSTANT CONTROL RELAY MODULE |
| FOG | 10 | FOG LAMPS |
| HAZARD | 15 | ELECTRONIC FLASHER, POWERTRAIN CONTROL MODULE (PCM) |
| AIR COND. | 15 | A/C-HEATER, ANTI-LOCK BRAKES, CONSTANT CONTROL RELAY MODULE (CCRM) |
| HORN | 15 | HORN, SHIFT LOCK |
| DOOR LOCK | 30 | POWER DOOR LOCKS |
| METER | 10 | REVERSING LAMPS, REMOTE KEYLESS ENTRY (RKE), INSTRUMENT CLUSTER, REAR WINDOW DEFROST, SHIFT LOCK, WARNING CHIME |
| RADIO | 5 | EXTERIOR REAR VIEW MIRRORS, INTEGRATED CONTROL PANEL (ICP) |
| P. WINDOW (C. B.) | 30 | POWER WINDOWS |
| AIR BAG | 10 | JOINT CONNECTOR #12, AIR BAG DIAGNOSTIC MONITOR |
| ROOM | 10 | RAP SYSTEM, RADIO, COURTESY LAMPS, STARTING SYSTEM, WARNING CHIME, INTEGRATED CONTROL PANEL (ICP) |
| FUEL INJECTOR | 10 | HO2S, EVAPORATIVE EMISSION PURGE FLOW SENSOR |
| STOP | 20 | STOPLAMPS, DEACTIVATOR SWITCH |
| TAIL | 15 | PARK LAMP RELAY |
| WIPER | 20 | WIPER/WASHER, BLOWER MOTOR RELAY |
| ROOF OPENING | 15 | ROOF OPENING PANEL |

| AMPERE RATING | HOUSING COLOR |
|---|---|
| **BLADE TYPE FUSE** | |
| 4 | Pink |
| 5 | Tan |
| 10 | Red |
| 15 | Light Blue |
| 20 | Yellow |
| 25 | Natural |
| 30 | Light Green |

**POWER DISTRIBUTION**

The Generator, Battery and Starter Motor are connected through the 100 Amp Main Fuse in the Battery Junction Box. Other circuits originate from this Main Fuse and are protected by additional fuses in the Battery Junction Box and Central Junction Box.

The fuses in the Battery Junction Box are powered at all times; as are the Blower, Door Lock, Fog, Hazard, Horn, Radio, Room, Stop and Tail Fuses in the Central Junction Box, the Ignition Switch and the Main Light Switch. The other fuses are powered through the Ignition Switch or the Main Light Switch.